

LIST:

MULTIPLE DATATYPE:

```
n_list=[2.9,3000,'hello',"welcome to python list",True]
```

```
print(n_list)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

```
[2.9, 3000, 'hello', 'welcome to python list', True]
```

NESTED LIST:

```
n_list=["happy",[2,0,1,5]]
```

```
print(n_list[0][3])
```

```
print(n_list[1][2])
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

```
p
```

```
1
```

NEGATIVE INDEXING:

```
n_list=['p','r','o','b','e']
```

```
print(n_list[-1])
```

```
print(n_list[-5])
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

```
e
```

```
p
```

LIST USING APPEND() AND EXTEND():

```
odd=[1,3,5]
```

```
odd.append(7)
```

```
print(odd)
```

```
odd.extend([9,11,13])
```

```
print(odd)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

[1, 3, 5, 7]

[1, 3, 5, 7, 9, 11, 13]

+ OPERATOR AND * OPERATOR:

```
odd=[1,3,5]
```

```
print(odd+[9,7,5])
```

```
print(["re"]*3)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

[1, 3, 5, 9, 7, 5]

['re', 're', 're']

LIST LENGTH AND REMOVE:

```
thislist=["apple","banana","cherry"]
```

```
print(len(thislist))
```

```
thislist.remove("banana")
```

```
print(thislist)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

3

['apple', 'cherry']

LOOP FOR LIST:

```
thislist=["apple","banana","cherry"]
```

```
for x in thislist:
```

```
    print(x)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

apple

banana

cherry

WHILE LOOP:

```
thislist=["apple","banana","cherry"]
```

```
i=0
```

```
while i<len(thislist):
```

```
    print(thislist[i])
```

```
    i=i+1
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

apple

banana

cherry

SORT LIST:

```
thislist=["orange","mango","kiwi","pineapple","banana"]
```

```
thislist.sort()
```

```
print(thislist)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

['banana', 'kiwi', 'mango', 'orange', 'pineapple']

SORT DESCENDING:

```
thislist=["orange","mango","kiwi","pineapple","banana"]
```

```
thislist.sort(reverse=True)
```

```
print(thislist)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

['pineapple', 'orange', 'mango', 'kiwi', 'banana']

INSERT():

```
fruits=['apple','banana','cherry']
```

```
fruits.insert(1,"orange")
```

```
print(fruits)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

```
['apple', 'orange', 'banana', 'cherry']
```

REMOVE FOR POP():

```
thislist=["apple","banana","cherry"]
```

```
thislist.pop(1)
```

```
print(thislist)
```

```
thislist.pop()
```

```
print(thislist)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

```
['apple', 'cherry']
```

```
['apple']
```

DEL KEYWORD:

```
thislist=["apple","banana","cherry"]
```

```
del thislist[0]
```

```
print(thislist)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

```
['banana', 'cherry']
```

CLEAR LIST:

```
thislist=["apple","banana","cherry"]
```

```
thislist.clear()
```

```
print(thislist)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

```
[]
```

```
COUNT():
```

```
fruits=["apple","banana","cherry","apple"]
```

```
x=fruits.count("apple")
```

```
print(x)
```

```
OUTPUT:
```

```
===== RESTART:
```

```
C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====
```

```
2
```

```
LIST COPY():
```

```
list=[10,20,30,40]
```

```
new_list=list.copy()
```

```
print("original list:",list)
```

```
print("copied list:",new_list)
```

```
OUTPUT:
```

```
===== RESTART:
```

```
C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====
```

```
original list: [10, 20, 30, 40]
```

```
copied list: [10, 20, 30, 40]
```

```
INDEX():
```

```
lst=[10,20,30,40,50]
```

```
value=30
```

```
position=lst.index(value)
```

```
print("list:",lst)
```

```
print("value:",value)
```

```
print("index:",position)
```

```
OUTPUT:
```

```
===== RESTART:
```

```
C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====
```

```
list: [10, 20, 30, 40, 50]
```

```
value: 30
```

```
index: 2
```

BUILT IN FUNCTION:

```
nums=[3,41,12,9,74,15]

print(len(nums))

print(max(nums))

print(min(nums))

print(sum(nums))

print(sum(nums)/len(nums))
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/list.py =====

6

74

3

154

25.666666666666668

TUPLE:

```
my_data=("hi","hello","bye")

print(my_data)

my_data2=(1,2.8,"hello world")

print(my_data2)

my_data4=((2,3,4),(1,2,"hi"))

print(my_data4)
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/tuple.py =====

('hi', 'hello', 'bye')

(1, 2.8, 'hello world')

((2, 3, 4), (1, 2, 'hi'))

ACCESING TUPLE:

```
my_data=("hi","hello","bye")

print(my_data)

print(my_data[0])
```

```
print(my_data[2])
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/tuple.py =====

```
('hi', 'hello', 'bye')
```

```
hi
```

```
bye
```

negative indexing:

```
my_data=(1,2,"KEVIN",8.9)
```

```
print(my_data)
```

```
print(my_data[-1])
```

```
print(my_data[-3])
```

OUTPUT:

===== RESTART:

C:/Users/student114/AppData/Local/Programs/Python/Python314/tuple.py =====

```
(1, 2, 'KEVIN', 8.9)
```

```
8.9
```

```
2
```